



Planning for Disaster

AWS Academy Cloud
Architecting



Introduction

Planning for Disaster

Module objectives

This module prepares you to do the following:

- Identify strategies for disaster planning, including recovery point objective (RPO) and recovery time objective (RTO) based on business requirements.
- Identify disaster planning for Amazon Web Services (AWS) service categories.
- Describe common patterns for backup and disaster recovery (DR) and how to implement them.
- Use the AWS Well-Architected Framework principles when designing a disaster recovery plan.

Module overview

Presentation sections

- Disaster planning strategies
- AWS disaster recovery planning
- Disaster recovery patterns
- Applying AWS Well-Architected Framework principles to disaster planning

Knowledge checks

- 10-question knowledge check
- Sample exam question

Hands-on labs in this module

Guided lab

- Configuring Hybrid Storage and Migrating Data with AWS Storage Gateway S3 File Gateway

As a cloud architect planning for disasters:



- I need to design architectures that mitigate the risk of disaster, and also support timely recovery from disasters when they occur, so that I can help to minimize the impact of disasters on the business.
- I need to consider the organization's business needs so I can apply disaster recovery patterns that balance cost, data loss, and recovery time.



Disaster planning strategies

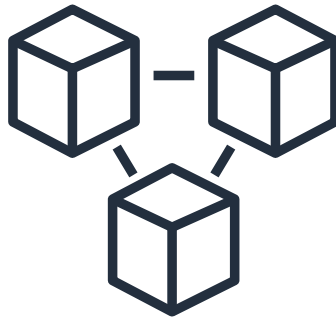
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Failures can occur at any scale



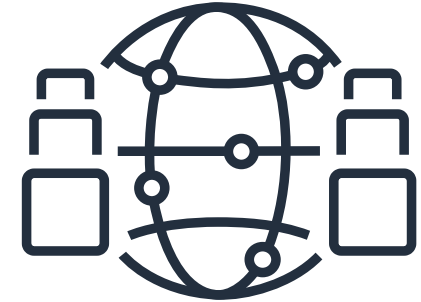
Small-scale events

A server stops responding.



Large-scale events

Multiple resources across an Availability Zone are unavailable.



Global events

Failure is widespread and impacts a large number of users and systems.

Avoiding and planning for disaster



Fault tolerance

Minimize how often your applications and data become unavailable.



Backup

Make sure you have a backup plan for handling data in case of a disaster.



Disaster recovery

Recover your data and get your applications back online after a disaster.

Factors that influence disaster planning strategies



Time dependency

How quickly does this need to be remedied to avoid business impact?

Data loss

What is the amount of data you are OK with losing?
What is the type of data you are OK with losing?

Geographic location

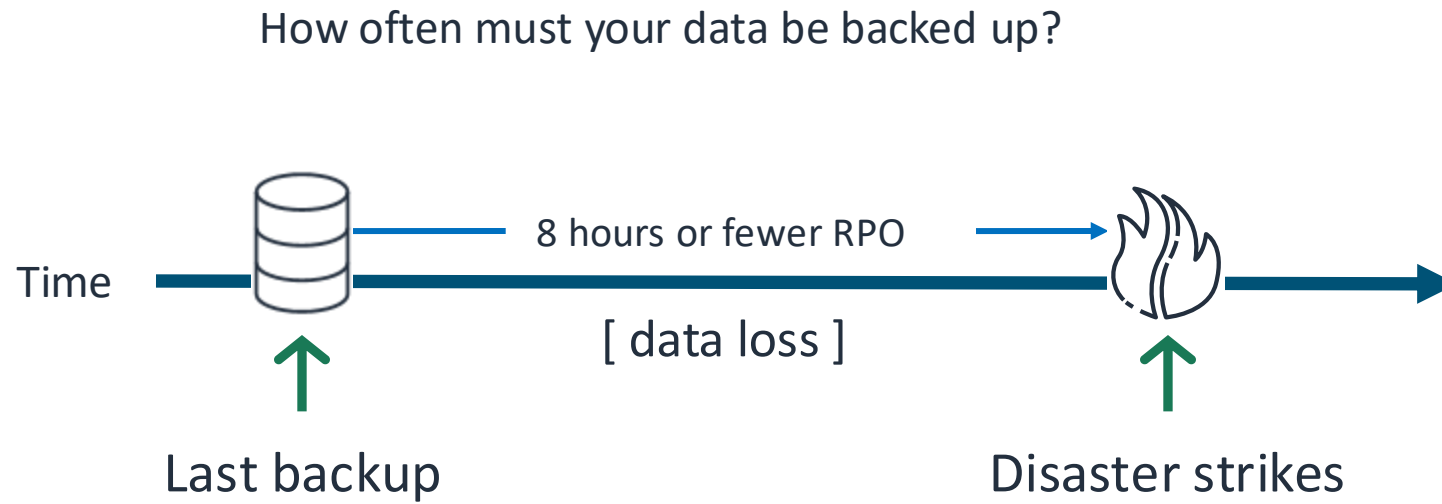
Does this impact multiple AWS Regions?
Do different Regions require different measurements of how quickly the must recover and how much data loss is OK?

Cost

Does the cost represent the correct level considering business impact and risk?

Determining RPO

RPO is the maximum acceptable amount of data loss, measured in time.



Example: Determining RPO



Determine that a loss of a maximum of 800 records is acceptable for your application.



→ Use existing patterns to determine that no more than 100 records are created an hour.

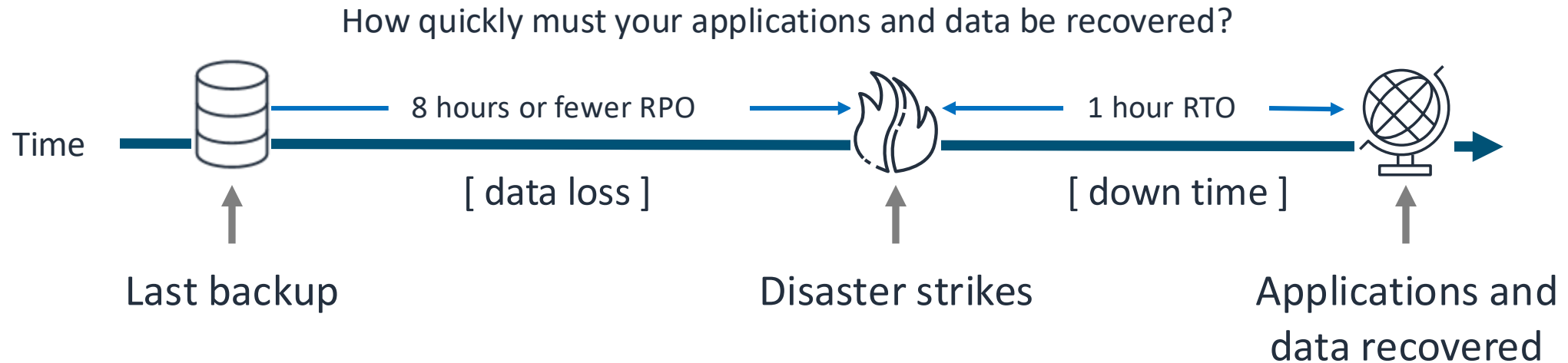


→ Calculate that an RPO of 8 hours (8*100 records) is acceptable.

Based on this information, if a disaster occurred at 10 p.m., the system should be able to recover all data that was in the system before 2 p.m.

Determining RTO

RTO is the maximum acceptable amount of time after a disaster strikes that a business process can remain out of commission.



Example: Determining RTO



Determine that a ticketing service for a local music venue can be restored within 2 hours.



→ The business calculates that after 2 hours of an outage, it will begin to lose revenue from lost sales.



→ Calculate that an RTO of 2 hours is acceptable.

Based on this information, if a disaster occurred at 9 p.m., the system should be returned to service before 11 p.m.

Preparing a BCP

A business continuity plan (BCP) is a system of prevention and recovery from potential threats to a company.

A BCP consists of the following:

- Business impact analysis
- Risk assessment
- Disaster recovery plan
- Evaluated and determined RPO and RTO

Key takeaways: Disaster planning strategies



- Failures can occur at any time and on a small, large, or global scale.
- A disaster recovery plan will help limit business and customer impact when a disaster occurs.
- RPO is the maximum acceptable amount of data loss after an unplanned data-loss incident.
- RTO is the amount of time an application, system, and process can be down without causing significant damage to the business.
- A BCP is a system of prevention and recovery from potential threats to a company that includes RPO and RTO.



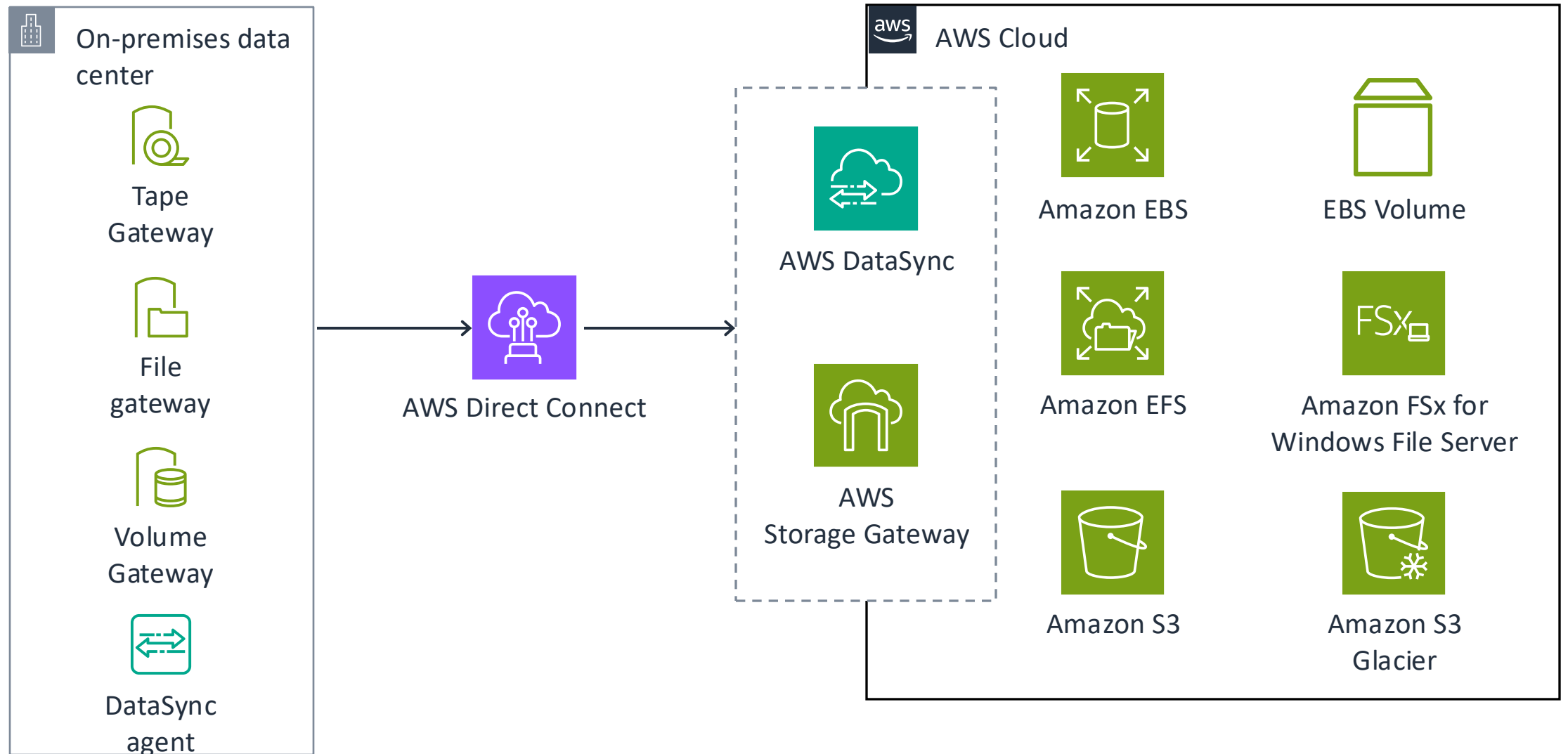
AWS disaster recovery planning

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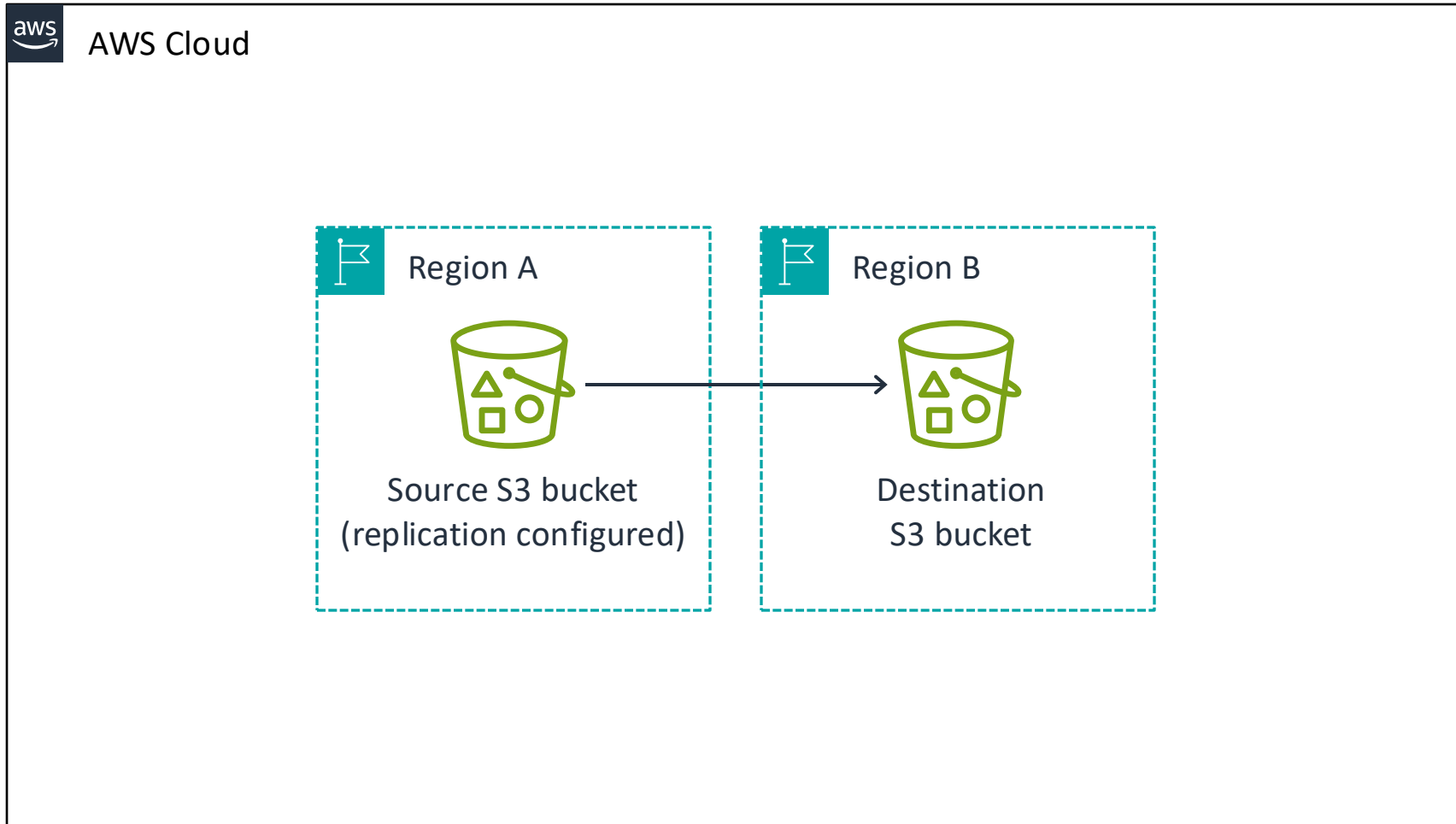
DR plans span more than one Region



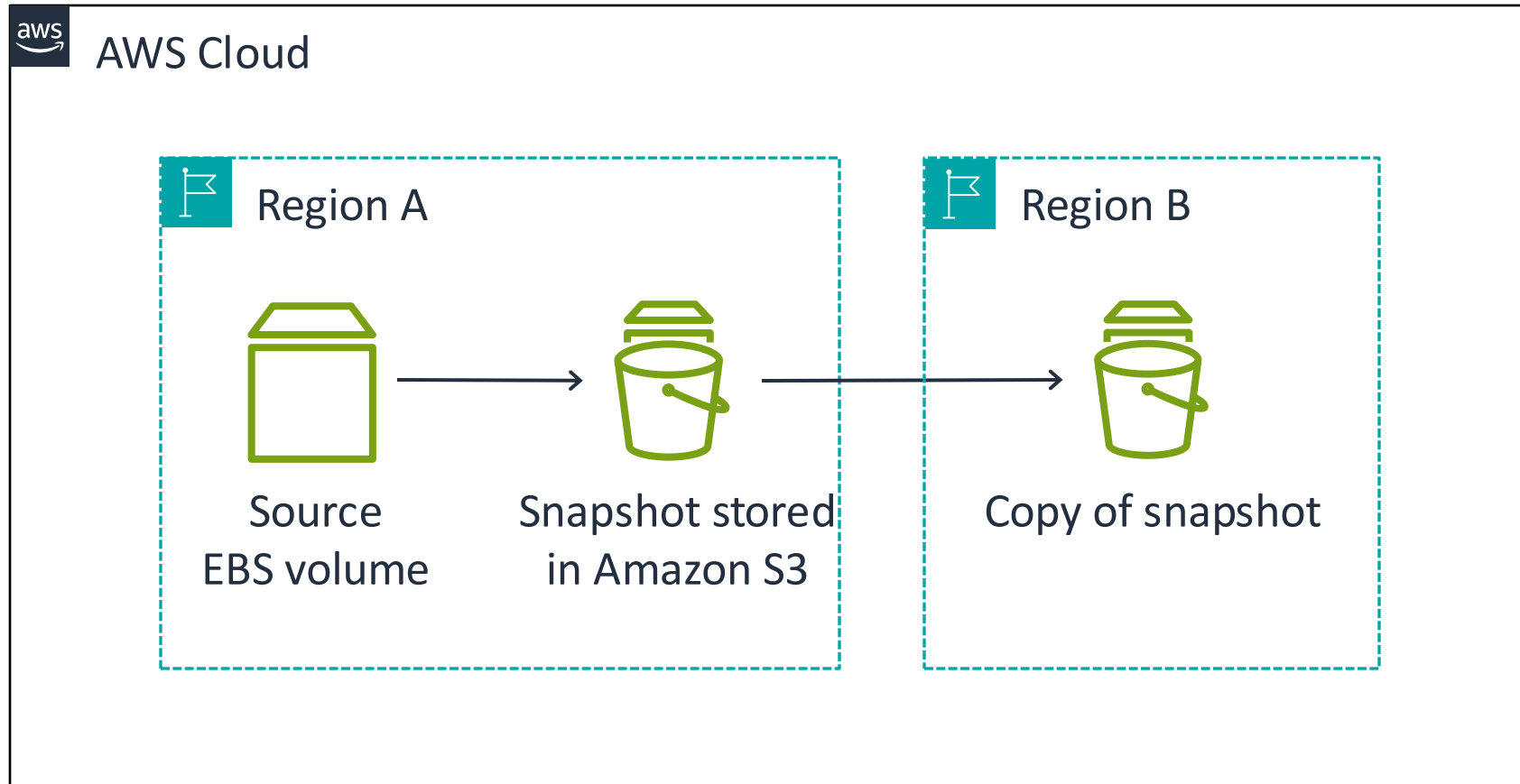
Storage and backup building blocks



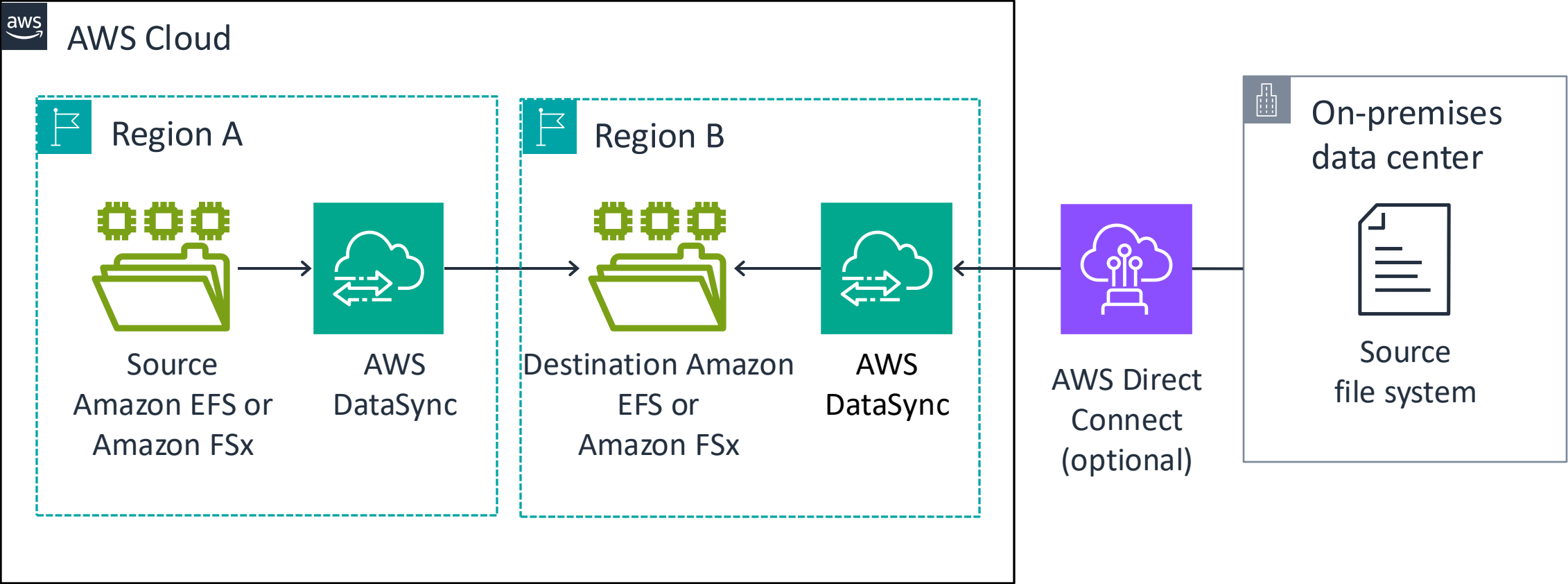
Configuring Cross-Region Replication



EBS volume snapshots

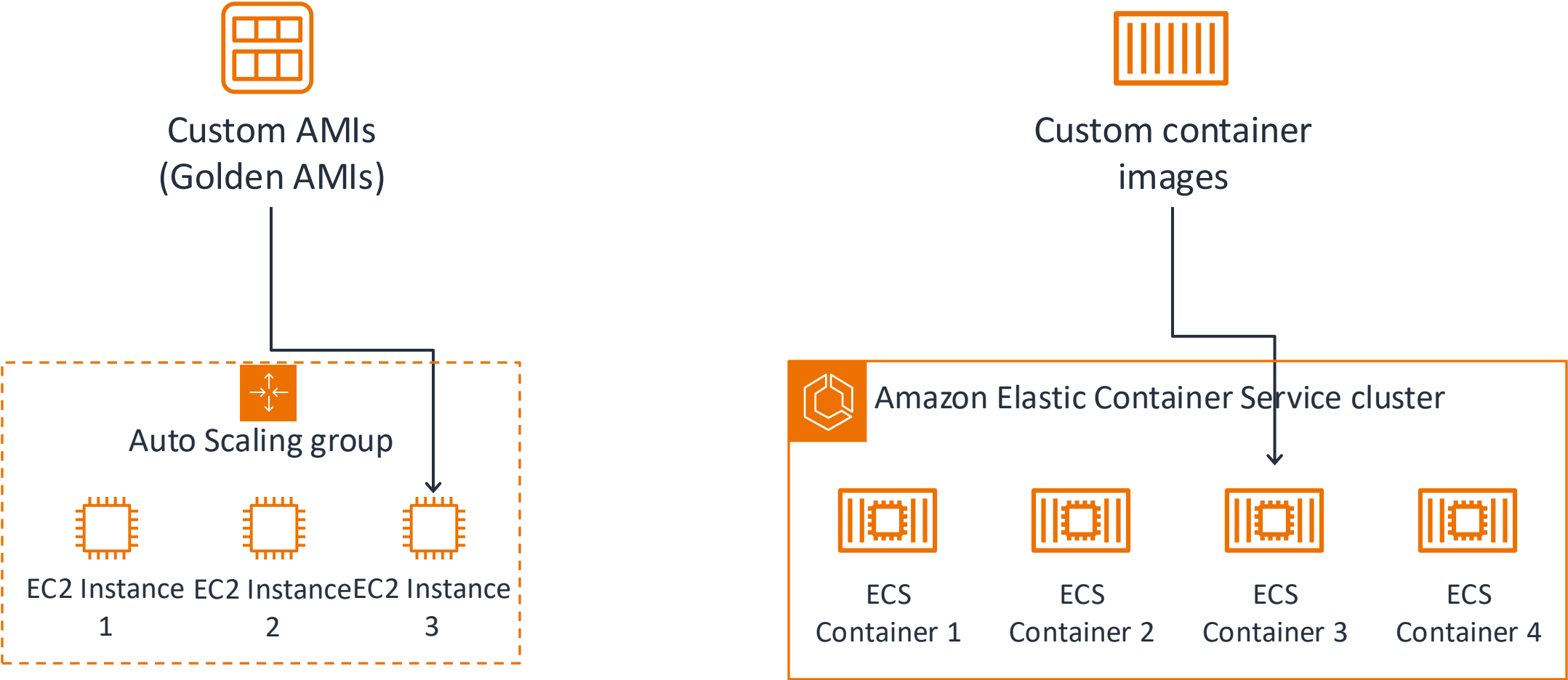


File system replication

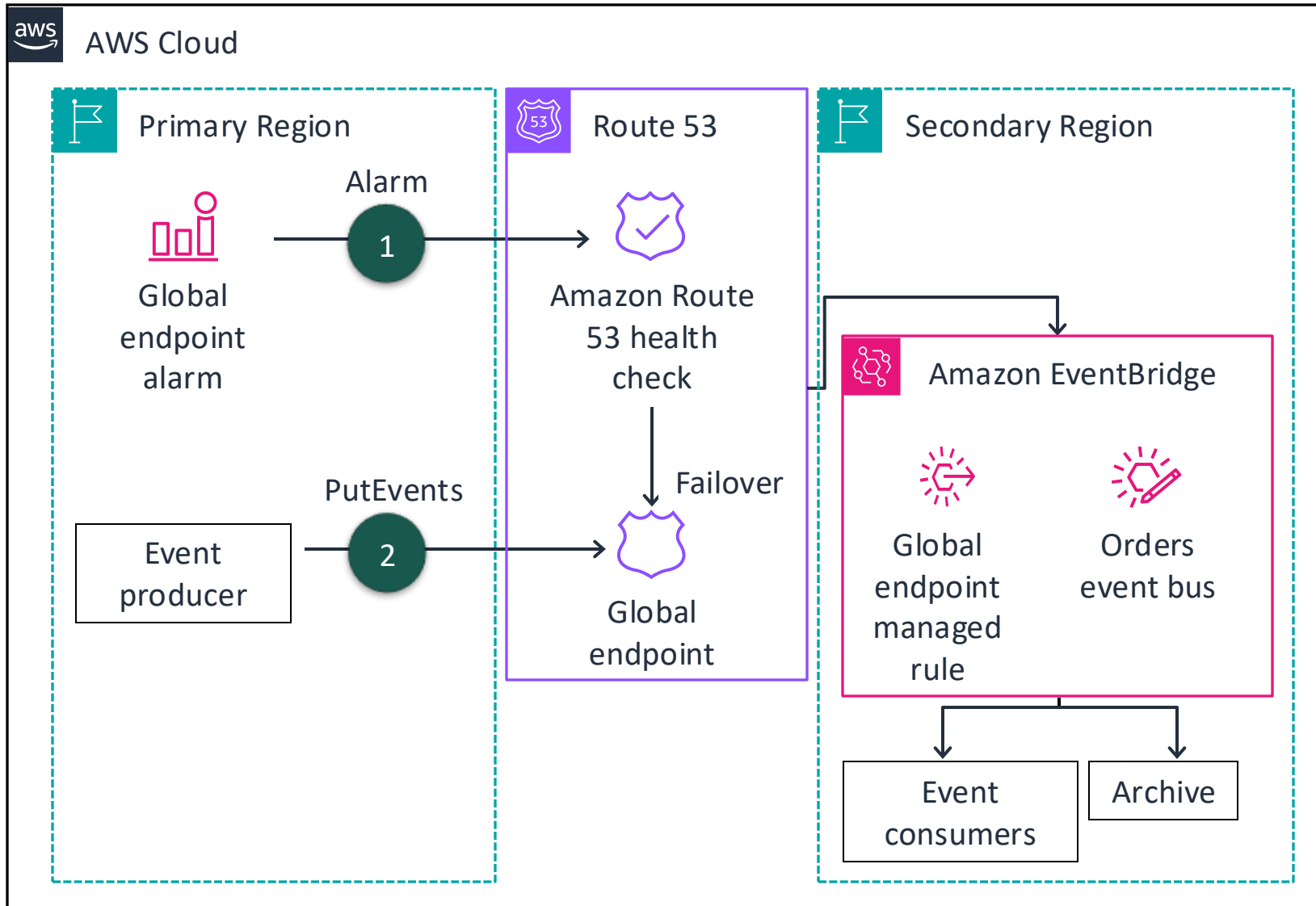


Recovering compute infrastructure

Obtain and boot new server instances or containers in minutes.



Harnessing EventBridge for regional failover

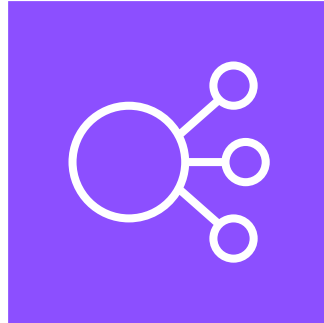


Designing for resiliency and recovery



Route 53

- Provides DNS-based load balancing
- Provides basic failover between endpoints or S3 websites



ELB

- Provides traffic distribution
- Makes disaster recovery implementation straightforward



Amazon VPN

Provides secure access to your on-premises network resources from your Amazon VPC through a VPN connection



AWS Direct Connect

Provides the dedicated network connection for fast, consistent data transfer between on-premises and AWS

Supporting database recovery



Amazon RDS

- Save a snapshot in a separate Region.
- Use read replicas and Multi-AZ deployments.
- Retain automated backups.



DynamoDB

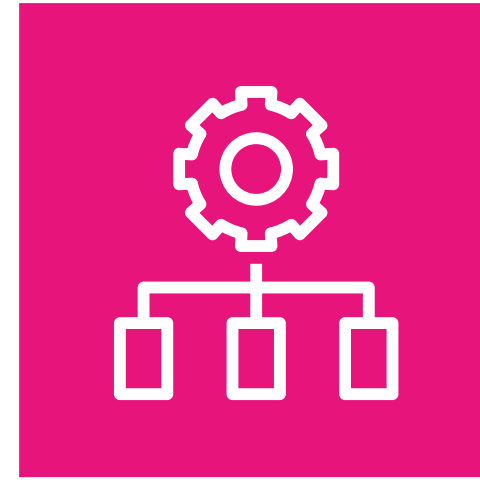
- Back up entire tables.
- Use point-in-time-recovery to restore tables.
- Create backups.
- Use global tables to build a multi-Region, multi-active database.

Replicating and redeploying environments



CloudFormation

- Use templates to quickly deploy collections of resources as needed.
- Duplicate production environments in a new Region or virtual private cloud (VPC) in minutes.



OpsWorks

- Manage and deploy applications across fleets.

Key takeaways: AWS disaster recovery planning



- Use features such as CRR, EBS volume snapshots, and Amazon RDS snapshots to protect data.
- Use networking features, such as Route 53 failover and ELB, to improve application availability.
- Use automation services, such as CloudFormation, as part of your DR strategy to quickly deploy duplicate environments when needed.
- Use EventBridge to use global endpoints to automatically restore data and failover to functioning servers.



Disaster recovery patterns

Planning for Disaster

Common disaster recovery patterns on AWS



Backup and restore

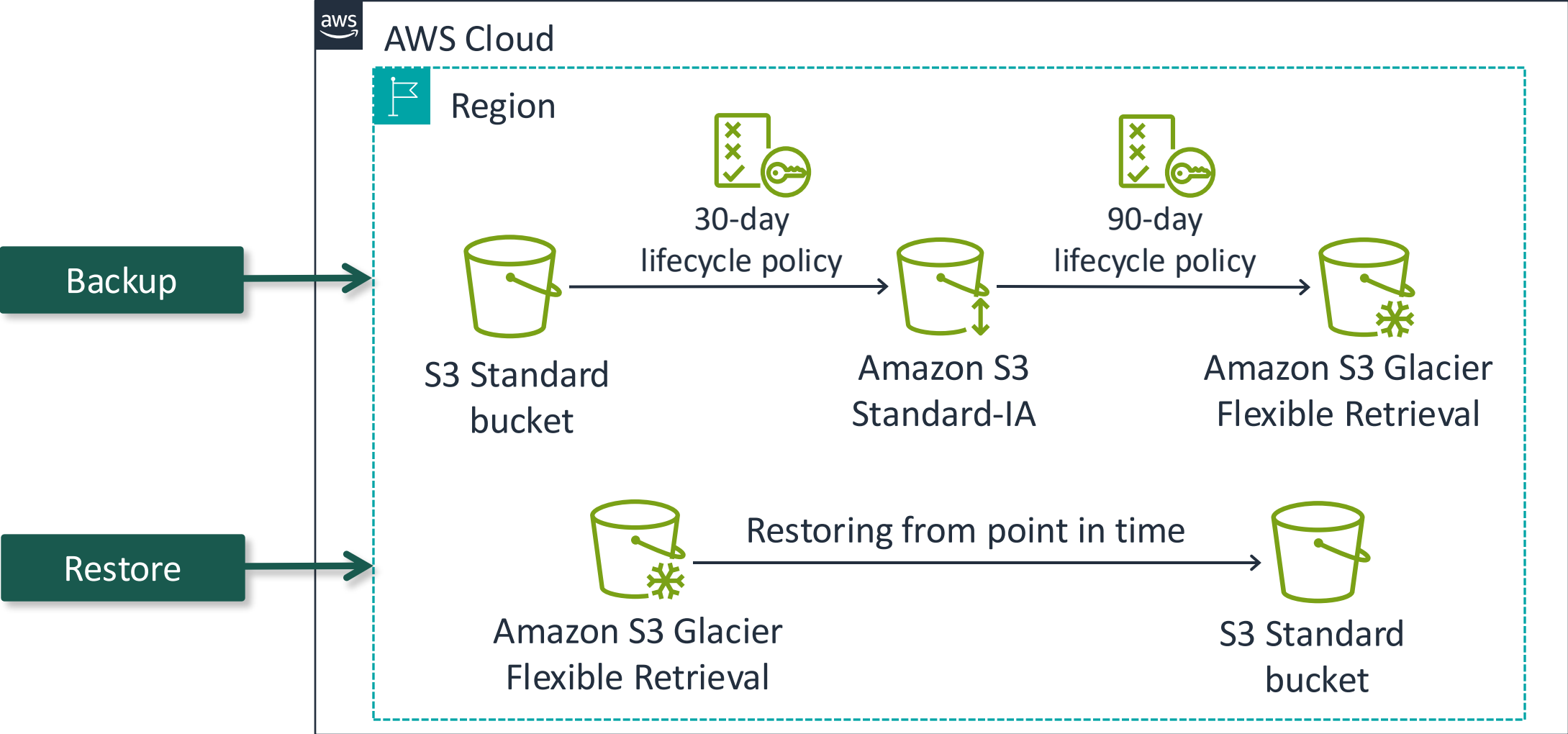
Pilot light

Warm standby

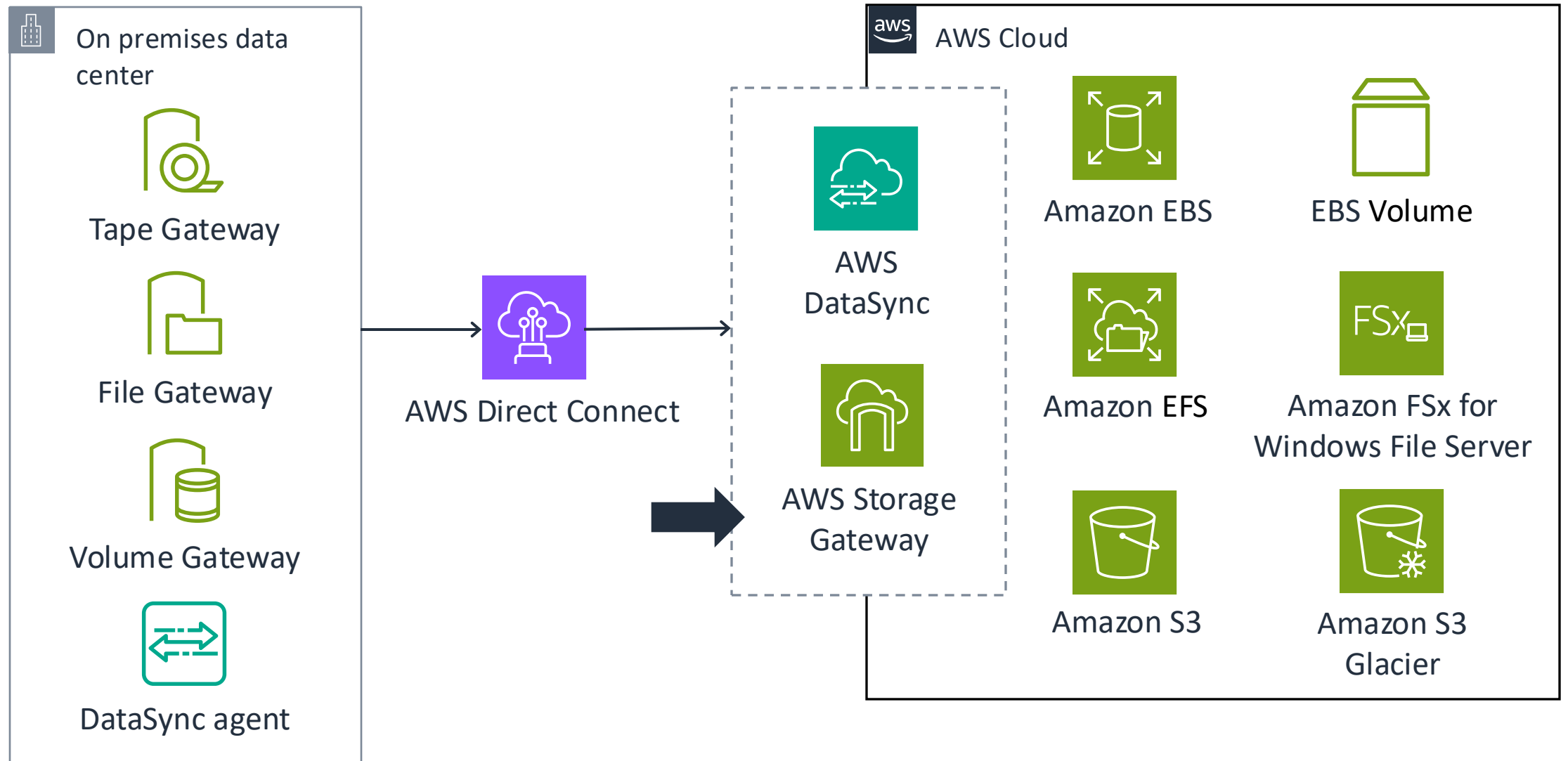
Multi-site

Each pattern is suited to a different combination of RPO, RTO, and cost-effectiveness.

Backup and restore pattern



AWS Storage Gateway backup and restore pattern



Implementing backup and restore



Preparation phase

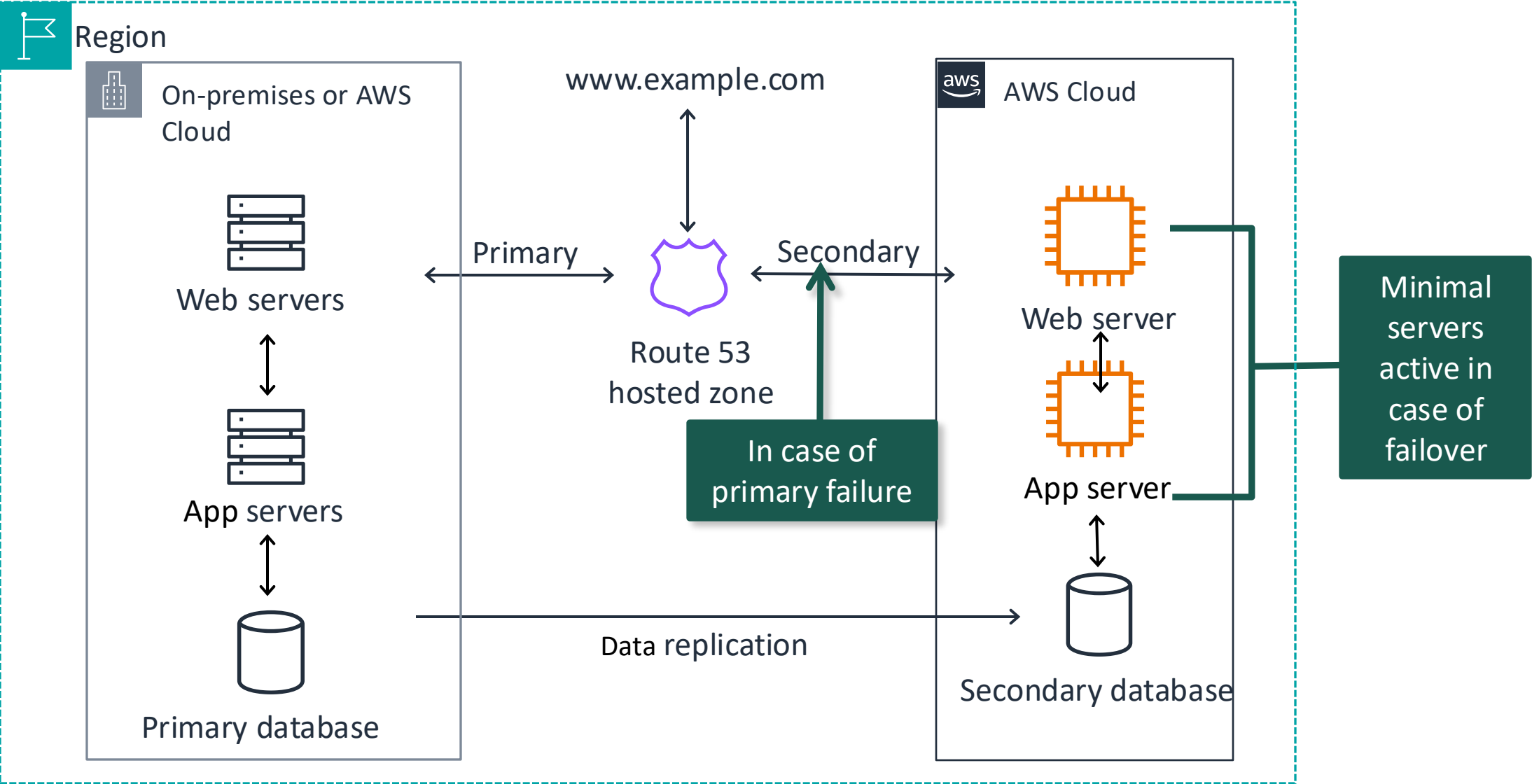
- Create backups of current systems.
- Store backups in Amazon S3.
- Document procedure to restore from backups.



In case of disaster

- Retrieve backups from Amazon S3.
- Restore required infrastructure.
- Restore system from backup.
- Route traffic to the new system.

Pilot light pattern



Implementing pilot light pattern



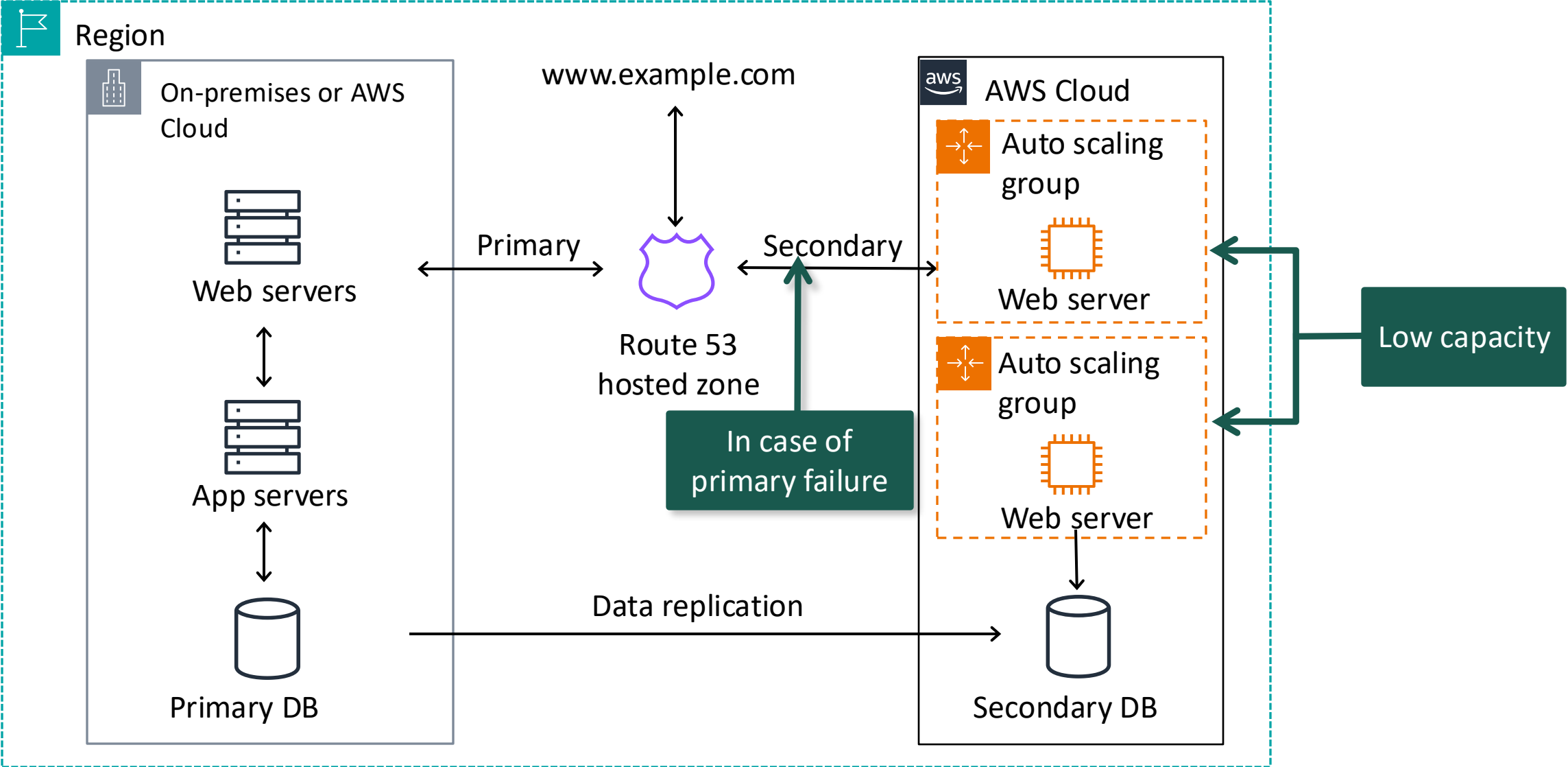
Preparation phase

- Configure EC2 instances to replicate servers.
- Create and maintain Amazon Machine Images (AMIs) of key servers where fast recovery is needed.
- Regularly run, test, and update these servers.

In case of disaster

- Bring up resources around the replicated core dataset.
- Then scale the system as needed to handle current production traffic.
- Switch over to the new system.

Warm standby pattern



Implementing warm standby pattern



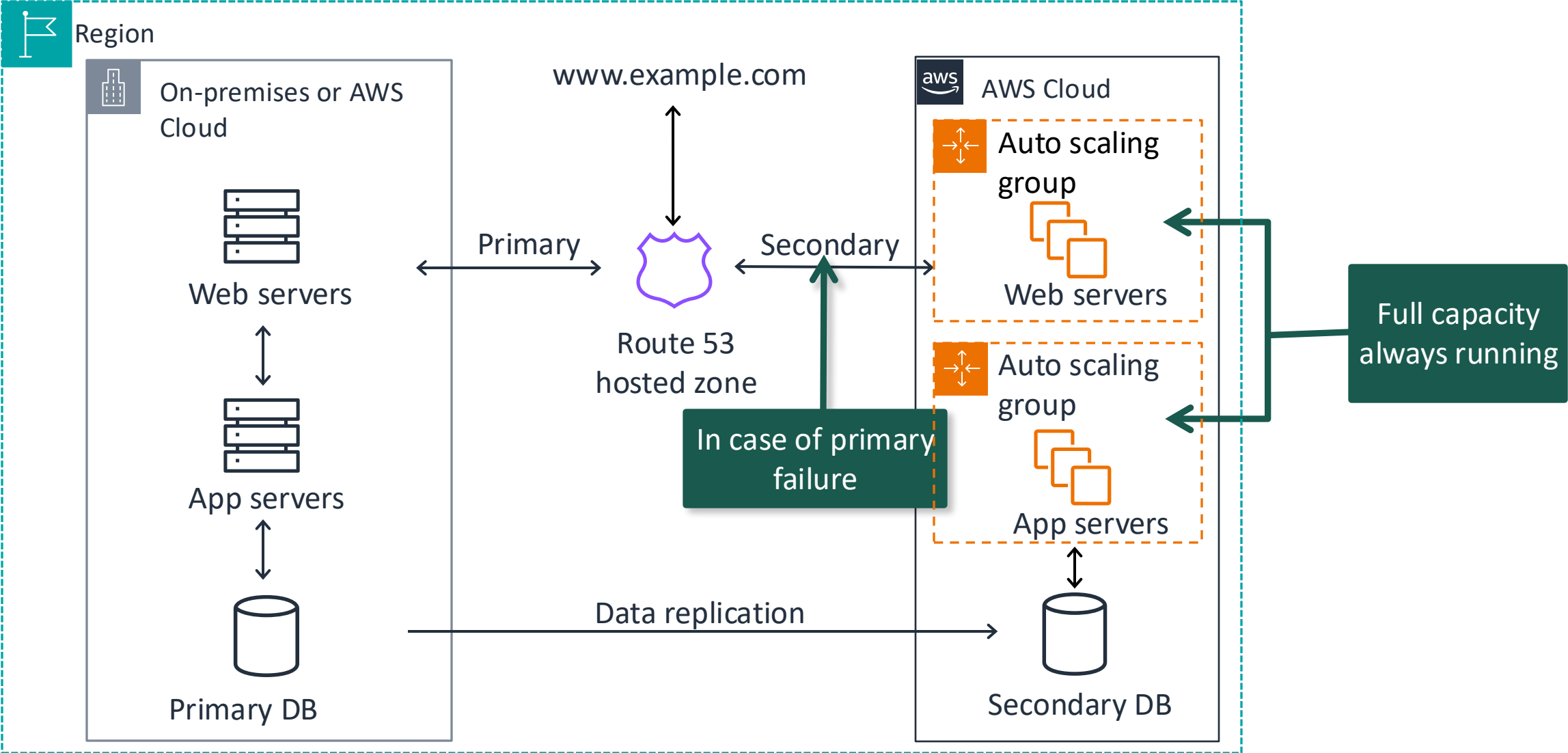
Preparation phase

- Is similar to pilot light
- Has all necessary components running 24/7, but not scaled for production traffic
- Need to conduct continuous testing on component

In case of disaster

- Failover most critical production load immediately
- Scale the system further to handle all production load (automatically)

Multi-site pattern



Implementing multi-site pattern



Preparation phase

- Similar to warm standby
- Configured for full scaling in or scaling out for production load
- Consider licensing cost of complete duplicate system



In case of disaster

Immediately failover all production load

Summary of common DR patterns



Backup and restore

Pilot light

Warm standby

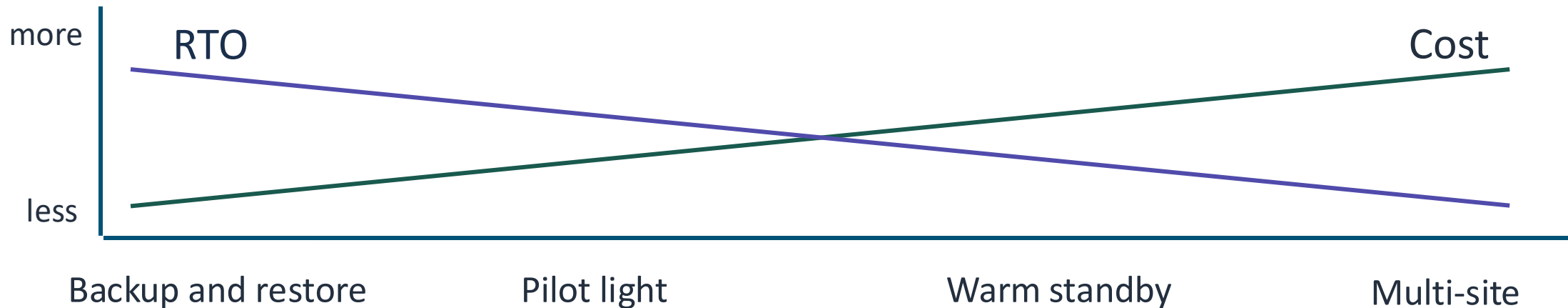
Multi-site

Cost

- Solutions that require RTO and RPO in hours
 - Lower-priority use cases
 - Solutions: Amazon S3, Storage Gateway
- Solutions that require RTO and RPO in 10s of minutes
 - Core services
 - Scale AWS resources
- Solutions that require RTO and RPO in minutes
 - Business-critical services
- Solutions that require RTO and RPO in near real-time
 - Automatic failover of environment in AWS to a running duplicate

Practice Game Day exercises

- Ensure that backups, snapshots, and AMIs are being created, and that they can be used to successfully restore data.
- Monitor your monitoring system.
- Establish RTO and RPO, and work to improve them where possible.



Key takeaways: Disaster recovery patterns



- Common disaster recovery patterns on AWS include backup and restore, pilot light, warm standby, and multi-site.
- Backup and restore is the most cost-effective approach. However, it has the highest RTO.
- Multi-site provides the fastest RTO. However, it costs the most because it provides a fully running production-ready duplicate.
- Storage Gateway provides three interfaces—File Gateway, Volume Gateway, and Tape Gateway—for data backup and recovery between on-premises and the AWS Cloud.



Guided lab: Configuring Hybrid Storage and Migrating Data with AWS Storage Gateway S3 File Gateway (AWS Storage Gateway lab)

AWS Storage Gateway lab tasks



- In this lab, you perform the following main tasks:
 - Configure an S3 File Gateway with an NFS file share and attach it to a Linux instance.
 - Migrate a set of data from the Linux instance to an S3 bucket.
 - Create and configure a primary S3 bucket to migrate on-premises server data to AWS.
 - Create and configure a secondary S3 bucket to use for cross-Region replication.
 - Create an S3 lifecycle policy to automatically manage data in a bucket.

Debrief: AWS Storage Gateway lab

- Which configuration do you need to enable in the source bucket and the destination bucket for cross-Region replication to work?
- How did you choose the correct AMI to use for the File Gateway appliance?



Applying AWS Well-Architected Framework principles to disaster planning

Planning for Disaster

Well-Architected Framework best practices for planning for disasters



Reliability



Operational
Excellence



Security

Best practice approach: Failure management—plan for disaster recovery



Reliability

Best practices

Define recovery objectives for downtime and data loss.

Use defined recovery strategies to meet the recovery objectives.

Test disaster recovery implementation to validate the implementation.

Best practice approach: Operate–manage workload and operations events



Operational
Excellence

Best practice

Define a customer communication plan
for outages.

Best practice approach: Identity and access management— permissions management



Security

Best practice

Establish an emergency access process.

Key takeaways: Applying AWS Well-Architected Framework principles to disaster planning



- Define recovery objectives for downtime and data loss based on business requirements.
- Use defined recovery strategies to meet the recovery objectives based on business requirements.
- Test disaster recovery implementation to validate the implementation.
- Define a customer communication plan for outages.
- Establish an emergency access process.



Module wrap-up

Planning for Disaster

Module summary

This module prepared you to do the following:

- Identify strategies for disaster planning, including RPO and RTO based on business requirements.
- Identify disaster planning for AWS service categories.
- Describe common patterns for backup and disaster recovery and how to implement them.
- Use the AWS Well-Architected Framework principles when designing a disaster recovery plan.

Considerations for the cafe

- Discuss how you, as a cloud architect, might advise the café, based on the key cloud architect concerns presented at the start of this module.



Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

Sample exam question

A solutions architect must create a disaster recovery (DR) solution for a company's business-critical applications. The maximum acceptable amount of data loss is 7 minutes. The DR solution also requires the deployment of a completely functional version of the applications to handle the majority of traffic immediately, and then scale up to full capacity over time. Which disaster recovery pattern would provide the most cost-effective solution?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- Business-critical
- Functional version of the applications
- Handle the majority of traffic immediately
- Cost-effective

Sample exam question: Response choices

A solutions architect must create a disaster recovery (DR) solution for a company's **business-critical** applications. The maximum acceptable amount of data loss is 7 minutes. The DR solution also requires the deployment of a completely **functional version of the applications to handle the majority of traffic immediately**, and then scale up to full capacity over time. Which disaster recovery pattern would provide the most **cost-effective** solution?

Choice	Response
A	Backup and restore
B	Pilot light
C	Warm standby
D	Multi-site

Sample exam question: Answer

The answer is C.

Choice	Response
C	Warm standby



Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.